

MedGemma 27B image-reading ablation — 2026-09-16

This is a model-screening research result, not clinical validation or evidence for autonomous diagnosis.

Research question

Does `medgemma:27b` use the AgentClinic-NEJM images specifically enough to justify a single-model architecture, or should the next prototype compare specialist medical image models?

The four paired conditions were:

- `context_only`: patient-actor proxy context, final task, and five choices.
- `context_image`: the same prompt plus the matched NEJM image.
- `context_mismatched`: the same prompt plus a deterministic wrong image selected to maximize available subtype overlap.
- `image_only`: final task, choices, and matched image without clinical history.

The proxy context excludes the original question's formal imaging/pathology report, but it is not clinician-redacted and can retain patient-visible findings. NEJM images are composite publication screenshots rather than native DICOM volumes or WSI data.

Complete 120-case result

All 480 responses were valid and scorable. Every case had all four conditions, and no mismatched-image condition reused its own image.

Condition	Correct / n	Accuracy	Wilson 95% CI
Proxy context only	78 / 120	65.0%	56.1%–72.9%
Context + matched image	75 / 120	62.5%	53.6%–70.6%
Context + mismatched image	80 / 120	66.7%	57.8%–74.5%
Image only	31 / 120	25.8%	18.8%–34.3%

- Matched image versus no image: −2.5 points; 2 helpful and 5 harmful flips; answer-change rate 9.2%; exact McNemar $p=0.453$.
- Matched versus mismatched image: −4.2 points; 3 helpful and 8 harmful flips; answer-change rate 12.5%; exact McNemar $p=0.227$.
- Image-only accuracy was not significantly above five-choice chance of 20% (one-sided exact binomial $p=0.072$).

Direct-diagnosis subset

Condition	Correct / n	Accuracy	Wilson 95% CI
Proxy context only	63 / 90	70.0%	59.9%–78.5%

Condition	Correct / n	Accuracy	Wilson 95% CI
Context + matched image	60 / 90	66.7%	56.4%–75.5%
Context + mismatched image	61 / 90	67.8%	57.6%–76.5%
Image only	21 / 90	23.3%	15.8%–33.1%

Matched images were –3.3 points versus no image and –1.1 points versus mismatched images. Image-only performance was consistent with chance ($p=0.25$).

Subtype screening

Subtype	n	Context	Matched	Wrong	Image only	Matched gain	Matched over wrong
physical	51	68.6%	62.7%	66.7%	25.5%	–5.9 pp	–3.9 pp
CT scan	19	42.1%	42.1%	52.6%	31.6%	0.0 pp	–10.5 pp
dermatology	16	93.8%	81.2%	81.2%	18.8%	–12.5 pp	0.0 pp
hist/path	13	76.9%	69.2%	69.2%	30.8%	–7.7 pp	0.0 pp
xray	12	58.3%	66.7%	66.7%	41.7%	+8.3 pp	0.0 pp
ophthalmology	11	81.8%	81.8%	90.9%	27.3%	0.0 pp	–9.1 pp

X-ray was the only adequately sized group with a positive matched-image gain and the highest image-only accuracy, so it is the best first specialist-model PoC. However, matched and mismatched images had identical X-ray accuracy, so this run did not demonstrate image specificity. Other subtype differences are negative or null and have wide uncertainty.

Decision

This evaluation rejects the claim that the current MedGemma 27B + NEJM screenshot pipeline has already demonstrated sufficient image-reading ability. Correct images did not improve accuracy, did not outperform closely matched wrong images, and image-only performance was statistically compatible with chance.

The result supports a specialist-model proof of concept, not immediate production integration. vLLM cannot replace that comparison because vLLM is an inference engine, not a visual model. The correct architecture experiment is:

1. MedGemma alone.
2. Specialist alone, blinded to answer choices.
3. Specialist structured findings passed to the same MedGemma reasoner.
4. Clinician-verified oracle visual findings passed to MedGemma, to measure the maximum value the images could add.

Adopt a specialist only if the hybrid shows a predeclared paired improvement on a clinician-redacted, image-essential private holdout. A reasonable research gate is an overall gain of at least 5 percentage points with a paired 95% interval above zero, or at least 10 points in a priority modality replicated on an

independent cohort, with no increase in harmful flips or critical visual misses. These are study-design gates, not clinical acceptance thresholds.

Candidate sequence

- Broad updated baseline: MedGemma 1.5 4B, then MedSigLIP for classification/retrieval.
- Chest X-ray: MAIRA-2 and CheXagent.
- CT/MRI: MedGemma 1.5 4B on native volumes; RadFM as a research comparator.
- Whole-slide pathology: SlideChat or TITAN only after obtaining native WSI data.

The local host currently has no vLLM installation or specialist VLM weights. Its 16 GB GPU already requires CPU offload for the 17 GB Q4 MedGemma 27B model, so vLLM deployment of 27B vision weights requires a separate memory and parity study.

Official model and runtime references:

- [MedGemma model card](#) and [MedGemma 1.5 4B](#)
- [MedSigLIP 448](#)
- [MAIRA-2](#) and [CheXagent](#)
- [RadFM](#), [SlideChat](#), and [TITAN](#)
- [vLLM supported-model documentation](#)

Limitations

- Proxy context was not manually redacted by clinicians.
- Images are public, possibly present in model pretraining data.
- Many are multi-panel publication composites; subtype tags are overlapping and not primary-modality annotations.
- The experiment measures five-choice case reasoning, not visual finding sensitivity, localization, laterality, report quality, calibration, or clinical safety.
- Most subtype sample sizes are insufficient for model procurement or deployment decisions.